



ST BENEDICT'S

SUBJECT	PHYSICAL SCIENCE – NOV – P1
GRADE	11
EXAMINER	Mr Hilder
MODERATOR	Mr Ntombela
DATE	03 November 2025
MARKS	125
DURATION	2 hours

%

MARKING GUIDELINES

Positive Marking: Penalise the first error, but mark all subsequent steps based on the student's incorrect value, provided the principle is correct.

Units: Penalise one mark for the entire question if units are consistently omitted or incorrect in final answers.

Rounding: Accept answers rounded to two decimal places.

e&s → correct equation with substitution attempt

☒ – mark not awarded for, but – 1 mark if ...

Question	1	2	3	4	5	6	7	8	9	TOTAL
Mark										
Moderated Mark										
Question Total	18	23	15	18	16	17	5	7	16	

QUESTION 1

18 MARKS

1.1

1.1.1 The rate of change of position |OR| the rate of displacement or
|OR| the rate of change of displacement

|OR|

✓✓ (AON)

Rate of change of position

(2)

1.1.2 d-e ✓✓

(2)

1.1.3 c and f and i ✓✓

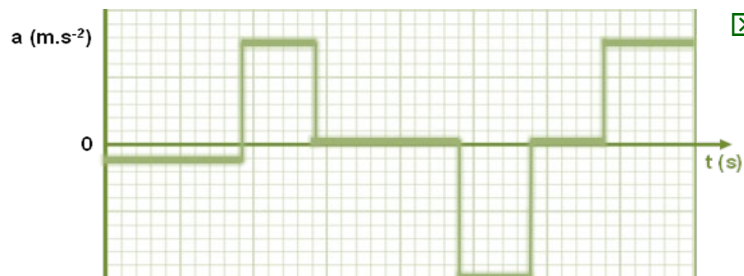
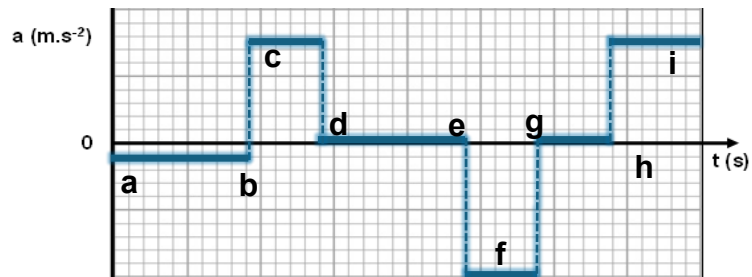
(2)

1.1.4 {e – g |OR| e – f |OR| f – g }✓✓

It has the steepest gradient.✓

(3)

1.1.5 ✓ - hi < cd & ✓ eg < ab & ✓ de and gh = 0 & ✓ labels



☒ inconsistent sign convention (-1)

(4)

1.2

$$v_c = \frac{\Delta x}{\Delta t}$$

$$1,2 = \frac{2}{\Delta t}$$

$$\Delta t = \frac{2}{1,2}$$

$$v_c = \frac{\Delta x}{\Delta t}$$

$$1,2 = \frac{2}{\Delta t} \quad \text{✓ eq. \& sub}$$

$$\Delta t = \frac{2}{1,2}$$

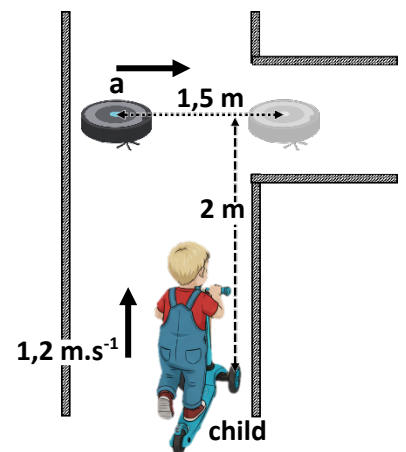
$$\Delta t = 1,67 \text{ s} \quad \text{✓}$$

$$s_R = ut + \frac{1}{2} a_R t^2$$

$$1,5 = (0)(1,67) + \frac{1}{2} a (1,67)^2 \quad \text{✓}$$

$$3 = 2,776 a$$

$$a = \boxed{1,08 \text{ m.s}^{-2}} \quad \text{✓ right ✓}$$



QUESTION 2

23 MARKS

2.1 Change in position.✓✓

(2)

2.2

(3)

$$\Delta y = v_i t + \frac{1}{2} g t^2$$

$$h = (4,2_{\pm})(1,57) + \frac{1}{2}(-9,8_{\pm})(1,57)^2 \checkmark_{e\&s}$$

$$h = \boxed{-5,48 \text{ m} \checkmark}$$

∴ h is 5,48 m above the trampoline

2.3

(4)

$$v_f^2 = v_i^2 + 2g\Delta y$$

$$0 \checkmark = 4,2^2 + 2(-9,8)(y_{peak} - 5,48 \checkmark_{coe 2.2}) \checkmark_{(eq \& sub)}$$

$$-17,64 = -19,6(y_{peak} - 5,48)$$

$$0,9 = y_{peak} - 5,48$$

$$y_f = \boxed{6,38 \text{ m} \checkmark}$$

2.4 Method A

Method 2

(3)

$$\Delta h = \frac{(v_i + v_f)}{2} \Delta t$$

$$0 - 5,48 \checkmark_{coe 2.2} = \frac{4,2 + v_f}{2} (1,57 - 0) \checkmark$$

$$v = \boxed{-11,18 \text{ m} \cdot \text{s}^{-1} \checkmark}$$

$$v_f = v_i + gt$$

$$v_f = 4,2 + (-9,8)(1,57) \checkmark \checkmark_{e\&s}$$

$$v_f = -11,86 \approx \boxed{11,19 \text{ m} \cdot \text{s}^{-1} \checkmark}$$

Method 3

$$v_f^2 = v_i^2 + 2a\Delta h$$

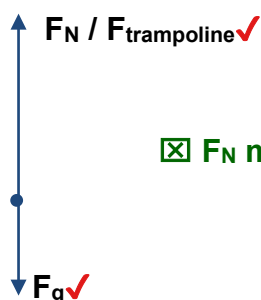
$$v_f^2 = 0 \checkmark_{zero} + 2(-9,8)(0 - 6,38 \checkmark_{coe 2.3})$$

$$v_f^2 = 125,048$$

$$v_f = \sqrt{125,048} = \boxed{\pm 11,182 \text{ m} \cdot \text{s}^{-1} \checkmark}$$

2.5

(2)



☒ F_N must be longer than F_g .

2.6

(4)

$$v_f^2 = v_i^2 + 2a\Delta y$$

$$0 = -11,2^2 + 2(a)(-0,8) \checkmark$$

$$a = 78,26 \text{ m} \cdot \text{s}^{-2} \checkmark$$

$$F_{\text{net}} = F_g + F_{\text{Trampoline}}$$

$$(70)(78,4) = (70)(-9,8) \checkmark + F_T$$

$$5478,2 = -686 + F_T$$

$$F_T = 6164,2 \text{ N} \checkmark$$

$$W_{\text{net}} = \Delta E_k$$

$$W_{\text{tramp}} + W_{F_g} = \Delta E_k \checkmark$$

$$F_{\text{tramp}}\Delta x + F_g\Delta x = 0 - \frac{1}{2}mv^2$$

$$-0,8F + (686)(-0,8) \checkmark = -\frac{1}{2}(70)(11,2^2)$$

$$-0,8F + 548,8 = -4382,56 \checkmark$$

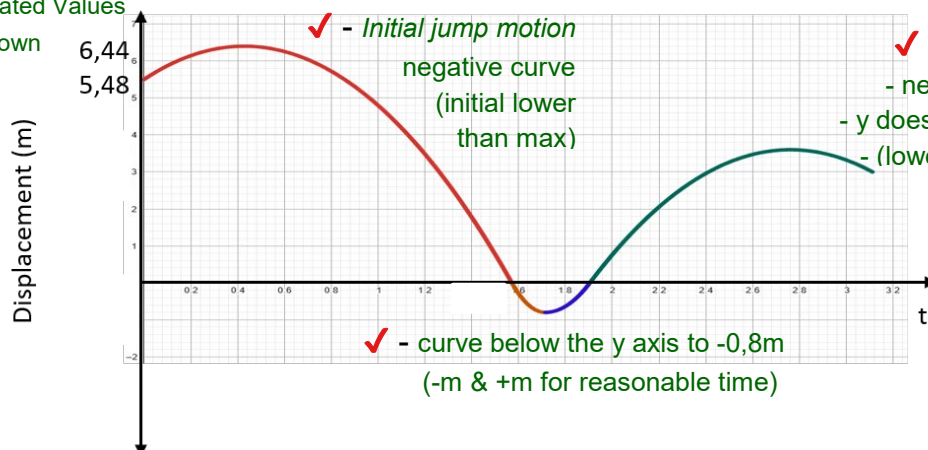
$$-0,8F = -4931,36$$

$$F = 6164,2 \text{ N} \checkmark$$

2.7

(5)

✓ -Calculated Values
shown

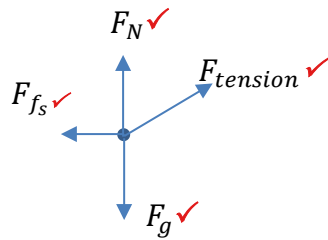


Axis labels (y and x)
units and names ✓
✗ any error

QUESTION 3

15 MARKS

3.1 Marking Criteria:



(✓) F_g / w / Weight: Arrow vertically down.

(✓) F_N / N / Normal Force: Arrow vertically up.

(✓) F_A / T / Tension: Arrow at 30° up and to the right.

(✓) f_s / f / Static Friction: Arrow horizontally to the left.

3.2 The **maximum** force...

...that opposes the tendency of motion of a stationary object. ✓

(2)

3.3

(3)

$$F_N = F_g - F_y$$

$$F_N = (70)(9.8) - \checkmark(300) \sin 30^\circ \checkmark$$

$$F_N = 686 - 150$$

$$F_N = \boxed{536 \text{ N}} \checkmark$$

3.4

(4)

$$F_{s\max} = \mu_s F_N$$

$$F_{s\max} = (0.6)(536) \checkmark$$

$$F_s^{\max} = 321.6 \text{ N} \checkmark$$

$$F_x = F_A \cos \theta$$

$$F_x = 300 \cos(30^\circ)$$

$$F_x = 267,30 \text{ N} \checkmark$$

The force he is applying is less than static friction so it won't move. ✓

3.5 By removing the cooler box the normal force acting on the canoe boat decreases. ✓

$$F_f = \mu_s F_N$$

$\therefore$ static friction is directly proportional to F_N ✓

(2)

QUESTION 4

18 MARKS

4.1 GREATER THAN ✓ (1)

4.2 Block B accelerates upwards, so there must be a **net (or resultant) force acting upwards** on it. ✓The forces on block B are weight (F_g) down and tension (F_T) up.For the net force to be up, F_T **must be greater than** F_g (OR $F_{net} = F_g - F_T = ma$). ✓ (2)

4.3 (4)

$$\begin{aligned}
 F_{net} &= ma \\
 F_{g\parallel} + F_T + F_f &= ma \\
 24,5 + F_T + (-2) &= 5a \\
 F_T &= 5a + 22,56 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 F_{net} &= ma \\
 F_g + F_T &= 2a \\
 -19,6 + F_T &= 2a \\
 F_t &= 2a - 19,6 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 F_{TAB} &= -F_{TBA} \checkmark \\
 5a + 22,56 &= -2a + 19,6 \\
 a &= -0,42 \text{ m} \cdot \text{s}^{-2}
 \end{aligned}$$

$$\begin{aligned}
 F_t &= 2a - 19,6 \\
 \therefore F_t &= \boxed{-20,44 \text{ N}} \checkmark
 \end{aligned}$$

4.4 LESS THAN ✓

The net force on the *entire system* ($F_{gB} - F_{g\parallel A}$) is accelerating the *total mass* not just m_B . ✓ |OR|The net force on the system is less than F_{gB} due to the opposing component of A's weight, so the acceleration must be less than g. ✓ (2)

4.5 The work done on an object by a net force is equal to the objects change in kinetic energy.

4.6 (3)

$$\begin{aligned}
 W &= \Delta E_k \checkmark \\
 F_{net} \Delta x &= E_{k_f} - E_{k_i} \\
 (F_{g\parallel} + F_f + F_t) \Delta x &= E_{k_f} - 0 \checkmark_{eq \& s} \\
 (24,5 + (-2) + (-20,44) \checkmark_{coe 4.3}) \Delta x &= 1,59 \\
 2,06 \Delta x &= 1,59 \\
 \Delta x &= \boxed{0,77 \text{ m}} \checkmark
 \end{aligned}$$

4.7

(5)

Constant velocity implies $F_{net} = 0$. ✓

$$F_{net} = ma$$

$$0 = F_{g_{parallel}} + F_f + F_{motor}$$

$$-F_{motor} = (5)(-9,8) \sin(30^\circ) + (-2) \text{ ✓}$$

$$F_{motor} = 24,245 \text{ N} \text{ ✓}$$

$$P_{output} = F_{motor}v$$

$$P_{output} = (24,245)(0,5) \text{ ✓}$$

$$P_{output} = \boxed{12,12 \text{ W}} \text{ ✓}$$

QUESTION 5

16 MARKS

5.1 The total linear momentum of an isolated system remains constant (OR is conserved).✓✓ (2)

5.2 (6)

Conservation of Momentum (Equation 1):

$$\begin{aligned}\Sigma P_{\text{initial}} &= \Sigma P_{\text{final}} \\ (m_A v_{Ai}) + (m_B v_{Bi}) &= (m_A v_{Af}) + (m_B v_{Bf}) \\ (0,25)(+2) + (0,5)(-5) &= (0,25)v_{Af} + (0,5)v_{Bf} \checkmark \\ 0,5 - 2,5 &= 0,25v_{Af} + 0,5v_{Bf} \\ -2 &= 0,25v_{Af} + 0,5v_{Bf} \checkmark \\ v_B &= 0,5v_A - 4\end{aligned}$$

Conservation of Kinetic Energy (Equation 2 OR Relative Velocity):

$$\begin{aligned}v_{Ai} - v_{Bi} &= v_{Bf} - v_{Af} \\ (+2) - (-5) &= v_{Bf} - v_{Af} \\ 7 &= v_{Bf} - v_{Af} \\ v_{Bf} &= 7 + v_{Af}\end{aligned}$$

$$\begin{aligned}\frac{1}{2} m v_{ai}^2 + \frac{1}{2} m v_{bi}^2 &= \frac{1}{2} m v_{af}^2 + \frac{1}{2} m v_{bf}^2 \\ \frac{1}{2} (0,25)(2)^2 + \frac{1}{2} (0,5)(5)^2 &= \frac{1}{2} (0,25)(v_a^2) + \frac{1}{2} (0,5)(v_b^2) \\ 0,5 + 6,25 &= 0,125v_a^2 + 0,25v_b^2 \\ 6,75 - 0,125v_a^2 &= 0,25v_b^2\end{aligned}$$

Substitution (sub v_{Bf} into Eq 1):

$$\begin{aligned}-2 &= 0,25v_{Af} + 0,5(7 + v_{Af}) \checkmark \\ -2 &= 0,25v_{Af} + 3,5 + 0,5v_{Af} \\ -2 - 3,5 &= 0,75v_{Af} \\ -5,5 &= 0,75v_{Af}\end{aligned}$$

$$\begin{aligned}6,75 - 0,125v_a^2 &= 0,25v_b^2 \\ 6,75 - 0,125v_a^2 &= 0,25(0,5v_a - 4)^2 \\ 6,75 - 0,125v_a^2 &= 0,25(0,5v_a - 4)(0,5v_a - 4) \\ 6,75 - 0,125v_a^2 &= 0,25(0,25v_a^2 - 2v_a - 2v_a + 16) \\ 6,75 - 0,125v_a^2 &= 0,25(0,25v_a^2 - 4v_a + 16) \\ \text{Type equation here.}\end{aligned}$$

$$\begin{aligned}v_{Af} &= \frac{-5,5}{0,75} = \boxed{-7,33 \text{ m.s}^{-1}} (7,33 \text{ m.s}^{-1}; \text{To the left}) \checkmark \\ v_{Bf} &= 7 + v_{Af} = 7 + (-7,33) = \boxed{-0,33 \text{ m.s}^{-1}} (0,33 \text{ m.s}^{-1}; \text{To the left}) \checkmark\end{aligned}$$

5.3 They will experience an equal ✓ magnitude of force. (NIIL)✓ (2)

5.4 The product of force and contact time. ✓✓ (2)

5.5 (3)

$$\begin{aligned}J &= F\Delta t \\ \Delta p &= F\Delta t \\ m(v_f - v_i) &= F\Delta t \checkmark \\ 0,25(-7,33 - 2) &= F(0,02) \checkmark \\ -1,8325 - 0,5 &= 0,02F \\ -2,3325 &= 0,02F\end{aligned}$$

$$F = \boxed{-116,63 \text{ N} \checkmark}$$

5.6 The magnetic collision attachments allow the trolleys to interact without physical contact $\checkmark$,

so there is minimal deformation, friction, sound or heat loss during the collision. $\checkmark$

This makes the collision closer to perfectly elastic. $\checkmark$

(3)

QUESTION 6

17 MARKS

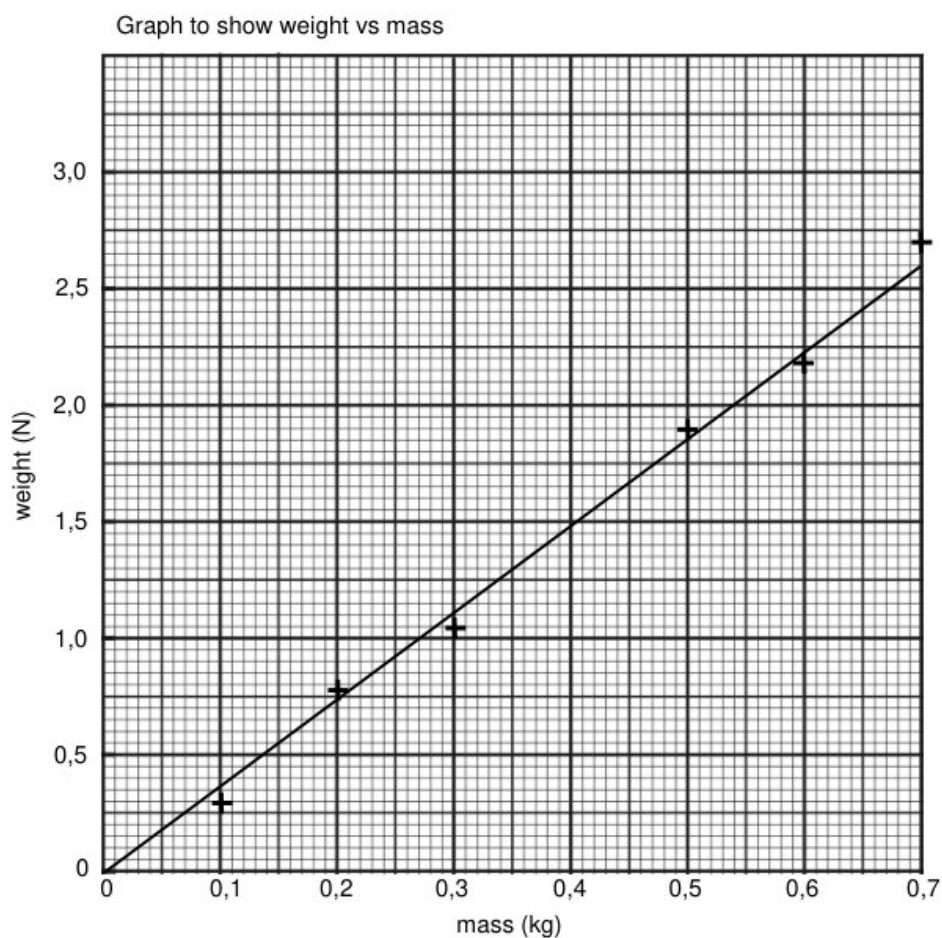
6.1

6.1.1 the force acting per unit mass✓✓ (2)

6.1.2 Y-axis label and scale ✓ (4)

5 points correctly plotted. [check 0,2 & 0,6]✓✓

L.O.B.F✓



6.1.3 (3)

$$\text{gradient} = \frac{\Delta y}{\Delta x}$$

$$\text{gradient} = \frac{y - \text{value from graph}}{x - \text{value from graph}} \quad \checkmark \quad \checkmark (\text{shown on graph})$$

$$\text{gradient} \approx 3,7 \text{ N} \cdot \text{kg}^{-1} \quad \checkmark (\pm 10\% \text{ answer})$$

6.1.4 From Graph: $\text{gradient} = \frac{F_g}{m}$ From equation: $F_g = mg$ (2)

$$\text{so: } g = \frac{F_g}{m} \quad \therefore \text{gravitational acceleration}$$

6.1.5 Mars✓ (1)

6.2

6.2.1 Positive✓ (1)

6.2.2 $Q_3 (+2\mu\text{C})$ experiences a net force to the left. Since Q_1 is positive, the force from Q_1 on Q_3 ($F_{1\text{on}3}$) is repulsive (to the right).✓ Therefore, Q_2 must exert a stronger attractive force ($F_{2\text{on}3}$) to the left on Q_3 for the net force to be left. For Q_2 to attract Q_3 , Q_2 must be negative✓

---OR---

If Q_2 was positive, both forces would be repulsive to the left, but then Q_1 would have to be negative ✓✓

---OR---

For F_{net} to be to the left and Q_1 to be positive and Q_3 to be positive, Q_2 must be negative, as positive Q_2 will give a stronger repulsive force from Q_2 to Q_3 ,✓ so it will not cause the net force to be left.✓ (2)

6.2.3 (4)

$$F = \frac{kQ_1Q_2}{r^2} \checkmark$$

$$0,012 = \frac{(9 \times 10^9)(2 \times 10^{-6})(Q_3)}{(1 + 1,5 \checkmark)^2} \checkmark_{\text{sub}}$$

$$F = \boxed{4,16 \times 10^{-6} \text{ C}} \checkmark$$

QUESTION 7**5 MARKS**

7.1 The green light will have a higher frequency and hence propagates more energy✓.

As wavelength is inversely proportional to frequency ($\lambda \propto \frac{1}{f}$)✓ (2)

7.2 (3)

$$E = \frac{hc}{\lambda}$$

$$E = \frac{(6,6 \times 10^{-34})(3 \times 10^8)}{530 \times 10^{-9}} \checkmark$$

$$E = 3,735 \times 10^{-19} J \checkmark$$

$$E(eV) = \frac{3,735 \times 10^{-19}}{1,6 \times 10^{-19}}$$

$$E = \boxed{2,33 \text{ eV}} \checkmark$$

QUESTION 8**8.1****(3)**

$$P = \frac{V^2}{R} \checkmark$$

$$4 = \frac{12^2}{R} \checkmark_{sub}$$

$$R = \boxed{36 \Omega} \checkmark$$

8.2 INCREASE $\checkmark$ **(1)****8.3** INCREASE $\checkmark$

The total current will decrease due to the increase in total resistance.

This means less voltage will be lost by the internal resistor. ($V_{int} = Ir$) $\checkmark$

So, the external voltage will increase over the 10W bulb.

$$(emf = V_{int} + V_{ext}) \checkmark$$

(3)**QUESTION 9**

9.1 An ohmic conductor is a component for which the current through it is directly proportional to the potential difference across it, $\checkmark$ provided the temperature remains constant. In other words, it obeys Ohm's Law and has a constant resistance. $\checkmark$ **(2)**

9.2**(3)**

$$V = IR \checkmark$$

$$3,2 = I(4) \checkmark$$

$$I = 0,8 \text{ A}$$

$$\therefore 0,8 \text{ A due to series circuit. } \checkmark$$

9.3**(4)**

$$emf = I(R_{ext} + r) \checkmark$$

$$emf = 0,8(12 \checkmark + 0,5 \checkmark)$$

$$emf = 10 \text{ V} \checkmark$$

9.4

$$emf = V_{ext} + V_{int}$$

$$10 = 8,8 + I(0,5)✓$$

$$1,2 = 0,5I$$

$$I = 2,4 A ✓$$

$$R_{ext} = \frac{V}{I}$$

$$R_{ext} = \frac{8,8}{2,4}$$

$$R_{ext} = 3,666 \Omega ✓$$

$$\frac{1}{R_{ext}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{3,666} = \frac{1}{12} + \frac{1}{R}$$

$$R = \boxed{5,28 \Omega} ✓$$

- 9.5** The internal resistance is consuming more power do to the greater current✓ and so converts electrical energy into thermal energy.✓ (2)